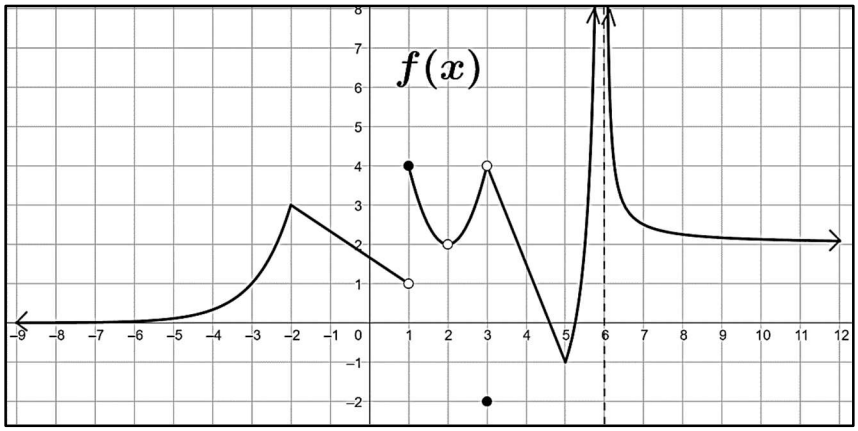
Worksheet 1-2 & 1-3 B: Limit Notation and Graphical Limits #2

AP Calculus Name _____



The graph of the function $f(x)$ is shown above and has a vertical asymptote at $x = 6$ and horizontal asymptotes of $y = 0$ and $y = 2$.

1. $\lim_{x\to 3} f(x)$

2. $\lim_{x\to 1^-} f(x)$

3. $\lim_{x\to 2^+} f(x)$

4. $\lim_{x\to -2} f(x)$

5. $\lim_{x\to 1} f(x)$

6. $f(1)$

7. $\lim_{x\to 5} f(x)$

8. $f(3)$

9. $\lim_{x\to 2} f(x)$

10. $\lim_{x\to 1^+} f(x)$

11. $f(2)$

12. $\lim_{x\to 3^+} f(x)$

For 13 – 16: We have not covered all of the concepts below, but they will be covered in more detail later in Unit 1. Use your current understanding of limits to see if you can find the following limit values!

13. $\lim_{x\to 6} f(x)$

4. $\lim_{x\to \infty} f(x)$

15. $\lim_{x\to -\infty} f(x)$

16. Find all values of c where $\lim_{x\to c} f(x) = 4$.